

Automated School IT Helpdesk Ticket Classification in ServiceNow

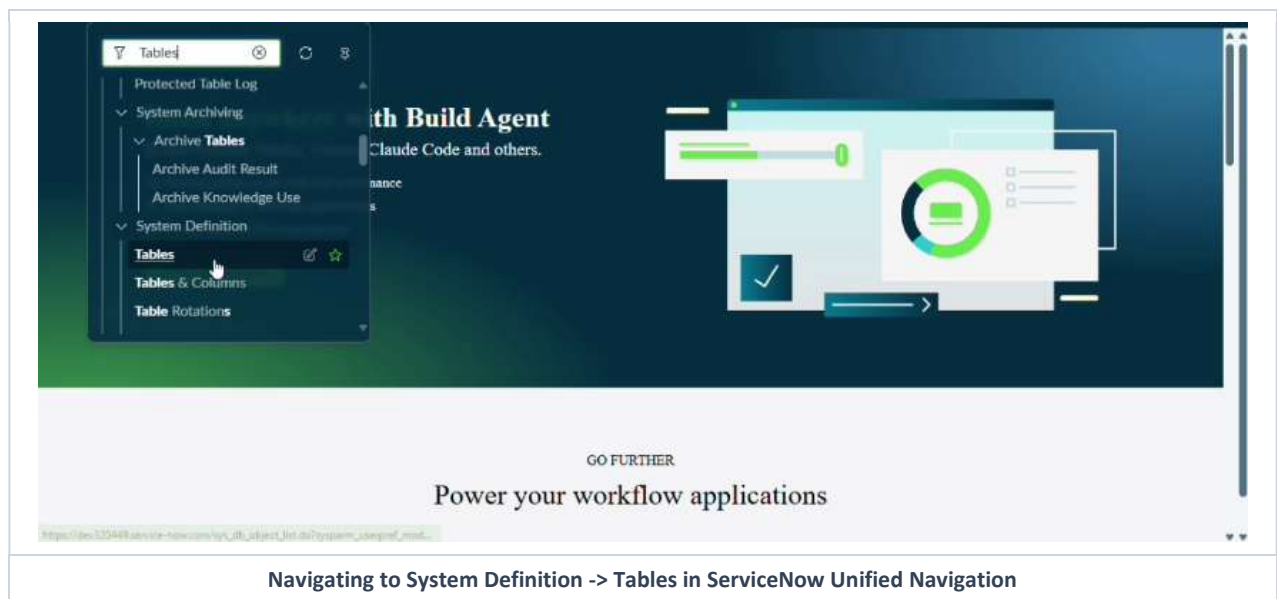
PHASE 5: PROJECT DEVELOPMENT PHASE

During development, the database schema, form layout, dependent dictionary attributes, and automated Flow Designer logic are constructed in the ServiceNow instance.

- **Table & Dictionary Configuration:** The Incident WorkFlow table is created under System Definition -> Tables with the Create module option unchecked and Auto-number enabled under the Controls tab. Fields and choice lists are arranged via Form Design. Opening Configure Dictionary (Advanced view) on the Subcategory field enables Dependent on field: Category, establishing backend value mappings: Network -> Wi-Fi, Hardware -> Projector, Access -> Forgot Password, and Performance -> Slow Computer.
- **Flow Construction:** In Process Automation -> Flow Designer, a new flow named Auto Classify School IT Tickets is configured to trigger when an Incident WorkFlow record is created with an empty Category field. Conditional If and Else If branches evaluate the Short description text for keywords (Wi-Fi/Network, Projector, Forgot password, and Slow Computer) to execute Update Record actions setting Category and Subcategory, followed by a Send Email action addressed to the caller's email data pill with the subject 'Your Request for the issue has been submitted'.

Activity 5.1: Custom Table Creation & Auto-Numbering (System Definition -> Tables)

1. Navigate to All -> System Definition -> Tables and click New.
2. Set Label = Incident WorkFlow (Backend Name auto-populates as u_incident_workflow) and uncheck Create module.
3. Switch to the Controls tab, check Auto-number, set Prefix = INC, Number = 1000, Number of digits = 7, and click Submit/Save.



< Table New record Submit Cancel

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Incident WorkFlow

* Name u_incident_workflow

Extends table

Application Global

Create module ☒

Create mobile module ☒

Add module to menu -- Create new --

New menu name Incident WorkFlow

Remote Table ☐

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

Creating the Custom Table 'Incident WorkFlow' (u_incident_workflow) in Global Scope

< Table New record Submit Cancel

* Label Incident WorkFlow

* Name u_incident_workflow

Extends table

Application Global

Create module ☐

Remote Table ☐

Columns Controls Application Access

Extensible ☐

Live feed ☐

Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format

Auto-number ☒

Prefix INC

Number 1000

Enabling Auto-Numbering (Prefix: INC, Starting Number: 1000) under the Controls Tab

The screenshot shows the 'Table Incident Workflow' configuration window. At the top, there are buttons for 'Delete', 'Update', and 'Delete All Records'. Below this, a message states 'Added new column Number'. A descriptive text box explains that a table is a collection of records. The configuration fields include:

- * Label: Incident Workflow
- * Name: u_incident_workflow
- Application: Global
- Remote Table: (empty)

 The 'Columns' tab is selected, showing options for 'Extensible' and 'Live feed', both of which are unchecked. A blue instruction bar says 'Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format'. Below this, the 'Prefix' is set to 'INC' and 'Number' is set to '1 / 000'.

Saved Table Definition with Access Control Role (u_incident_workflow_user) and Form Design Link

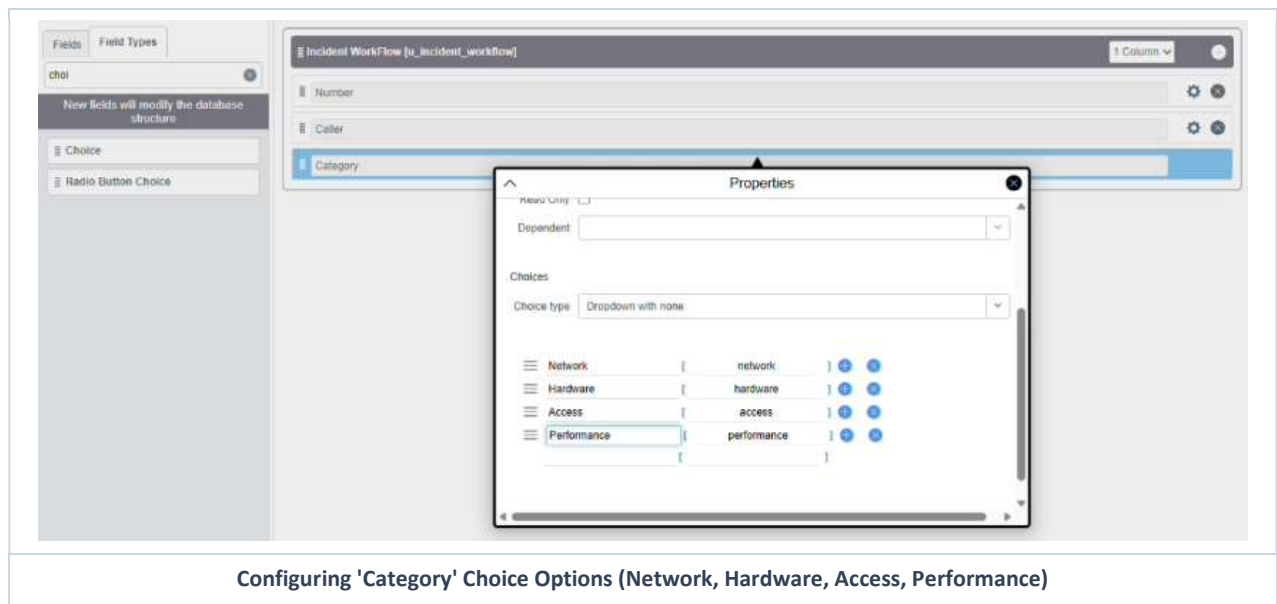
Activity 5.2: Form Design & Choice List Configuration

1. Open Form Design for Incident Workflow [u_incident_workflow].
2. Add Reference fields Caller (sys_user), Assigned Group (sys_user_group), and Assigned to (sys_user).
3. Add String fields Short Description and Description.
4. Add Choice fields Category, Subcategory, and State, and configure their dropdown options, then click Save.

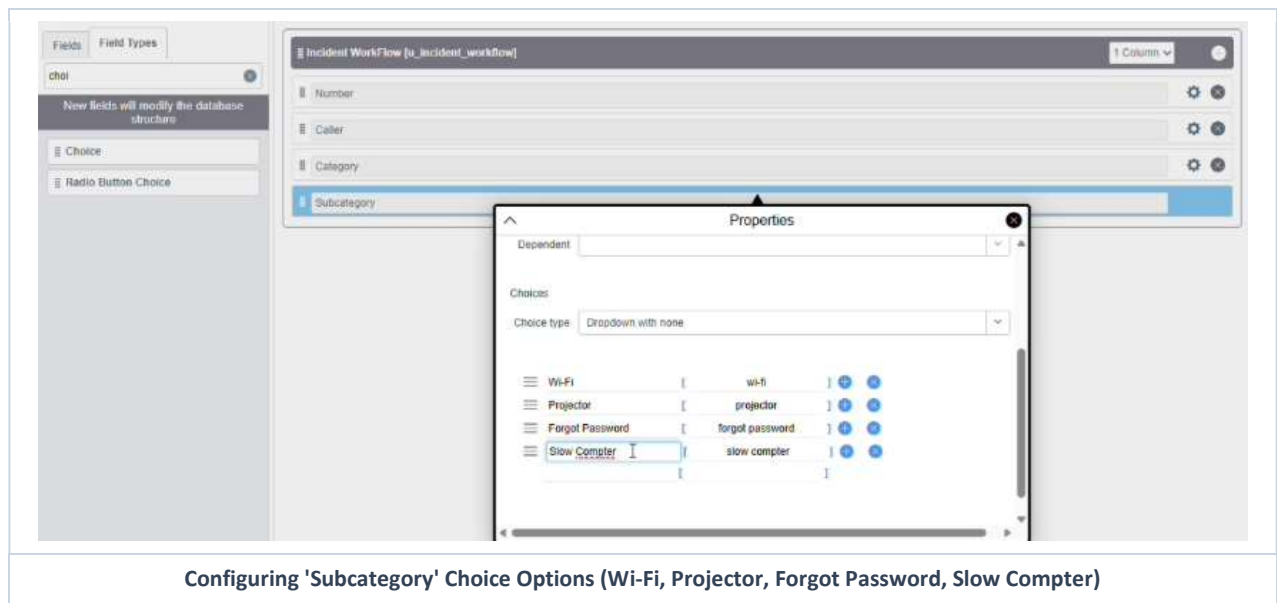
The screenshot shows the 'Form Design' interface for the 'Incident Workflow [u_incident_workflow]' table. On the left, a 'Fields' panel lists 'Reference' fields. The main area shows the 'Number' and 'Caller' fields. A 'Properties' dialog box is open for the 'Caller' field, with the following settings:

- Label: Caller
- Name: u_caller
- Type: Reference
- Reference: User [sys_user]
- Default: (empty)
- Mandatory: ☐
- Read Only: ☐
- Dependent: (empty)

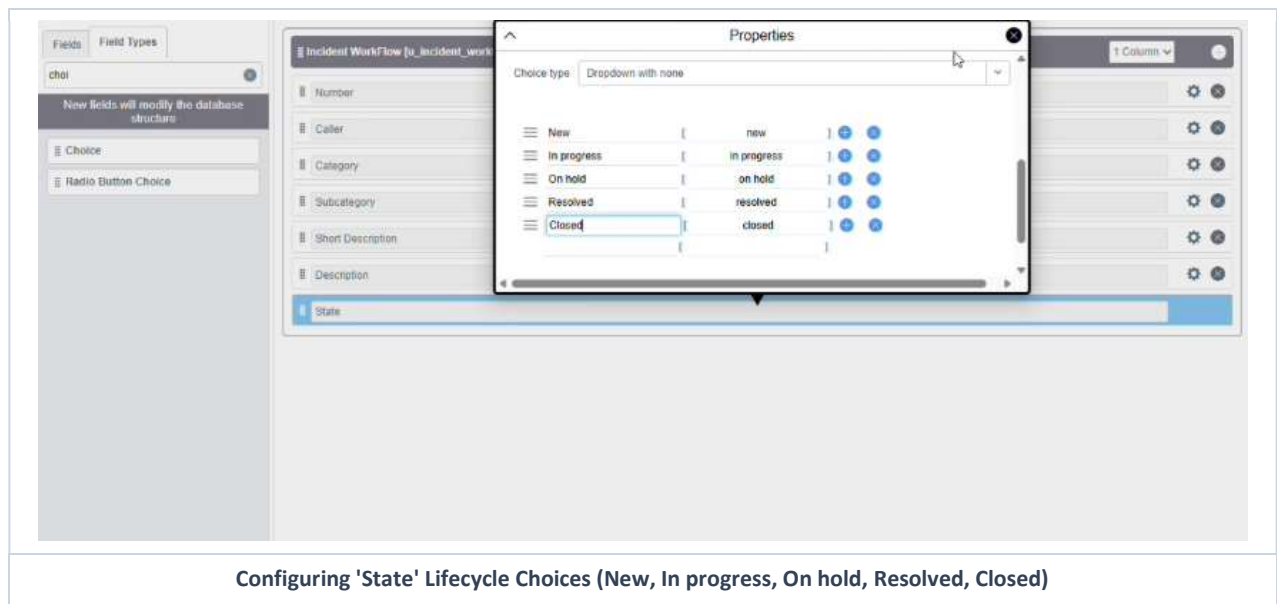
Configuring the 'Caller' (u_caller) Reference Field Pointing to User [sys_user]



Configuring 'Category' Choice Options (Network, Hardware, Access, Performance)

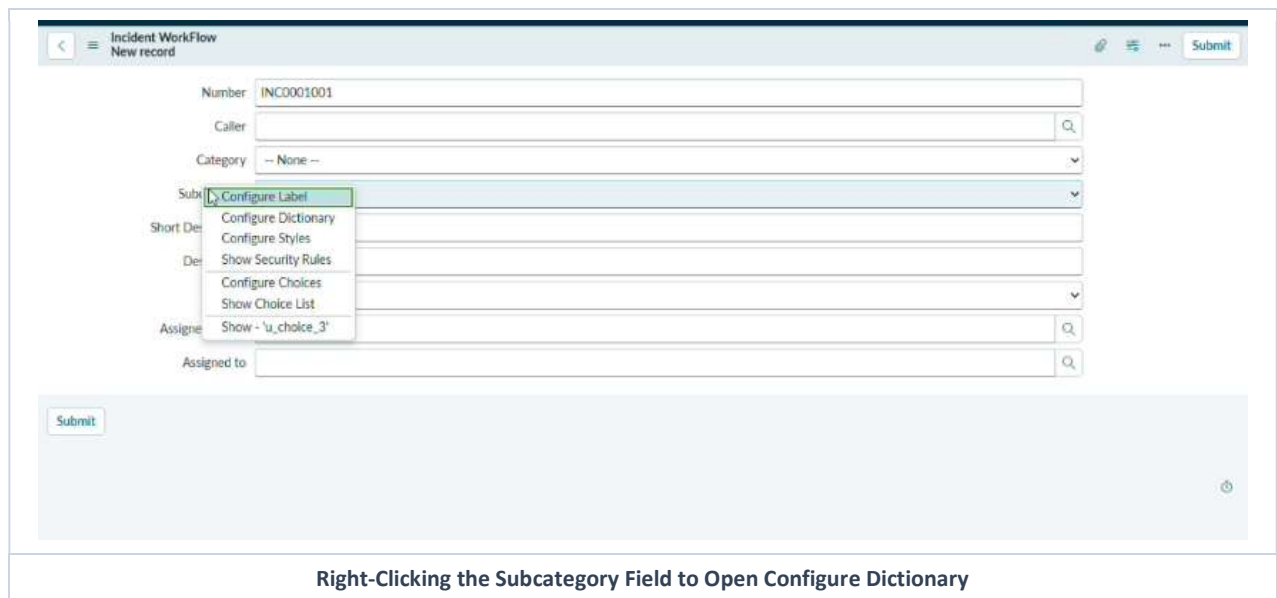


Configuring 'Subcategory' Choice Options (Wi-Fi, Projector, Forgot Password, Slow Compter)



Activity 5.3: Configuring Category-to-Subcategory Dictionary Dependency

1. Open the Incident Workflow form, right-click the Subcategory field label, and select Configure Dictionary.
2. Under Related Links, click Advanced view.
3. In the Dependent Field tab, check Use dependent field and set Dependent on field = Category.
4. In the Choices (4) related list at the bottom, populate the Dependent value column: Wi-Fi -> Network, Projector -> Hardware, Forgot Password -> Access, and Slow Computer -> Performance, then click Update.



Dictionary Entry Subcategory View: Advanced

Attributes: edge_encryption_enabled=true

Choice List Specification | **Dependent Field** | Calculated Value | Default Value

Specify a field whose value is used to determine choice list options for this field. [More info](#)

Use dependent field ☒

Dependent on field: Category

Create Choice List | Delete Column | Update

Related Links
[Show Table](#)
[Run Point Scan](#)
[Default view](#)

Access Controls | **Choices (4)** | Attributes (1) | Labels (1)

Enabling 'Use dependent field' on 'Category' in Dictionary Entry Advanced View

Dictionary Entry Subcategory View: Advanced

Use dependent field ☒

Dependent on field: Category

Create Choice List | Delete Column | Update

Related Links
[Show Table](#)
[Run Point Scan](#)

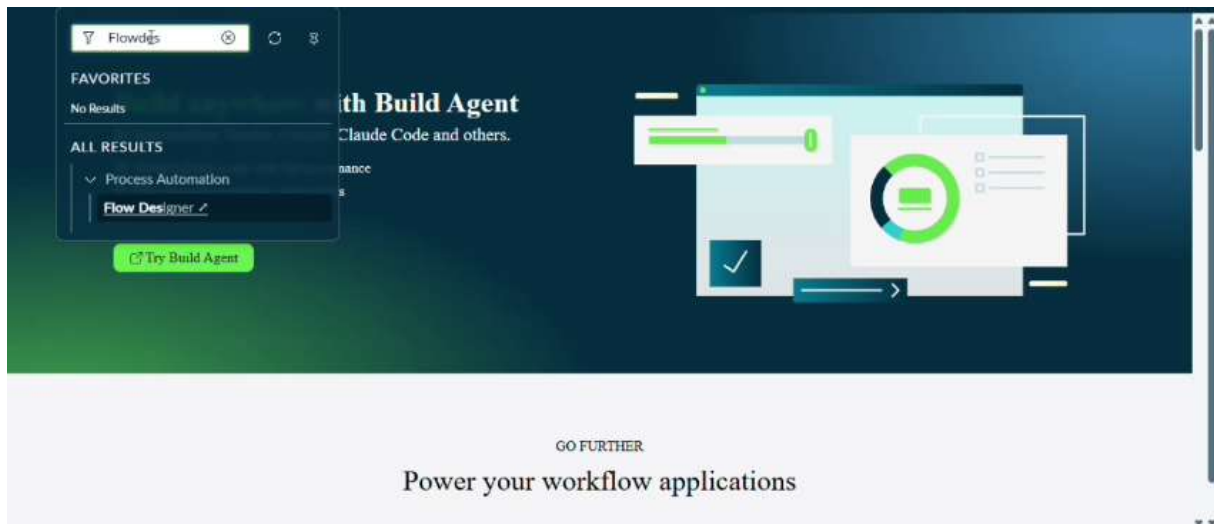
Access Controls | **Choices (4)** | Attributes (1) | Labels (1)

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
Forgot Password	forgot password	en	2	false	2026-09-23 01:57:01	Access
Slow Computer	slow computer	en	3	false	2026-09-23 01:57:01	Performance
Projector	projector	en	1	false	2026-09-23 01:57:01	Hardware
Wi-Fi	wi-fi	en	0	false	2026-09-23 01:57:01	Network

Mapping Dependent Category Values in the Subcategory Choices Related List

Activity 5.4: Flow Designer Automation & Email Notification Setup

1. Navigate to All -> Process Automation -> Flow Designer and click New -> Flow.
2. Enter Flow name = Auto Classify School IT Tickets in the Global application scope and click Build flow.
3. Configure the Trigger as Record -> Created on Table Incident Workflow [u_incident_workflow] with Condition: Category is empty.
4. Construct the 4 conditional classification branches (If / Else If) and the outbound Send Email action, then click Activate.



Opening Process Automation -> Flow Designer from the All Navigation Menu

The screenshot shows the 'Let's get the details for your flow' form. The form has the following fields: 'Flow name' (with a tooltip 'Name to uniquely identify your flow.') containing 'Auto Classify School IT Tickets', 'Application' (with a tooltip 'Application scope to create your flow in.') set to 'Global', and 'Description' (with a tooltip 'Describe your flow.') containing 'Describe your flow.'. A link '> Show additional properties' is below the description field. At the bottom right, there are 'Cancel' and 'Build flow' buttons.

Creating New Flow 'Auto Classify School IT Tickets' in Global Scope

Auto Classify School IT Tickets

Inactive

View

Test

Activate

TRIGGER

Incident WorkFlow Created where (Category is)

Trigger

Created

* Table

Incident WorkFlow (u_Incident_w...

Condition

Category

is

or

and

Add condition set

Advanced Options

Showing 1-5 of 5 (empty)

Network

Hardware

Access

Performances

Delete

Cancel

Done

Data

Collapse All

Flow Variables

Trigger - Record Created

Incident WorkFlow Record

Record

Incident WorkFlow Table

Table

Run Start Time UTC

Date/Time

Run Start Date/Time

Date/Time

Status: Modified | Application: Global

Configuring Flow Trigger — Record Created on Incident WorkFlow Where Category is Empty

Auto Classify School IT Tickets

Inactive

View

Test

Activate

TRIGGER

Incident WorkFlow Created where (Category is null)

ACTIONS

Select multiple

1

IF

Condition Label: SD is Wifi or Network

* Condition 1:

Trigger... Incident WorkFlow...

conta

Wi-Fi

or

and

OR

Condition 2:

Trigger... Short Descri...

conta

Network

or

and

Add another condition set(OR)

Data

Collapse All

Flow Variables

Trigger - Record Created

Incident WorkFlow Record

Record

Incident WorkFlow Table

Table

Run Start Time UTC

Date/Time

Run Start Date/Time

Date/Time

Status: Modified | Application: Global

Branch 1 (If) — Evaluating Short Description Contains 'Wi-Fi' OR 'Network'

Auto Classify School IT Tickets Inactive

1 If SD is Wifi or Network

2 then Update Incident WorkFlow Record

Action Properties

Action: Update Record

Action Inputs

* Record: Trigger... Incident WorkFlow...

* Table: Incident WorkFlow [u_Incident_w...]

Category: Network

Subcategory: Wi-Fi

Data Collapse All

- Flow Variables
 - Trigger - Record Created
 - Incident WorkFlow Record: Record
 - Incident WorkFlow Table: Table
 - Run Start Time UTC: Date/Time
 - Run Start Date/Time: Date/Time
 - 1 - If
 - 2 - Update Record
 - Incident WorkFlow Record: Record
 - Incident WorkFlow Table: Table
 - Action Status: Object

Status: Modified Application: Global

Branch 1 Action — Updating Incident WorkFlow Record to Category = Network, Subcategory = Wi-Fi

Auto Classify School IT Tickets Inactive

TRIGGER

Incident WorkFlow Created where (Category is null)

ACTIONS Select multiple

1 If SD is Wifi or Network

2 then Update Incident WorkFlow Record

3 Else if

Condition Label: SD is Projector or Hardware

* Condition 1: Trigger... Short Descri... conta Projector or and

Data Collapse All

- Flow Variables
 - Trigger - Record Created
 - Incident WorkFlow Record: Record
 - Incident WorkFlow Table: Table
 - Run Start Time UTC: Date/Time
 - Run Start Date/Time: Date/Time
 - 1 - If
 - 2 - Update Record
 - Incident WorkFlow Record: Record
 - Incident WorkFlow Table: Table
 - Action Status: Object
 - 3 - Else If

Status: Modified Application: Global

Branch 2 (Else If) — Evaluating Short Description Contains 'Projector'

Branch 2 Action — Updating Incident WorkFlow Record to Category = Hardware, Subcategory = Projector



Auto Classify School IT Tickets Inactive

7 * then Update Incident WorkFlow Record

Action Properties

Action: Update Record

Action Inputs

* Record: Trigger... Incident WorkFlow...

* Table: Incident WorkFlow [u_Incident_w...]

Category: -- None --

Subcategory: -- None --

+ Add field value

Wi-Fi
Projector
Forgot Password
Slow Computer

Delete Cancel Done

Status: Modified Application: Global

Data Collapse All

- Incident WorkFlow Record Record
- Incident WorkFlow Table Table
- Run Start Time UTC Date/Time
- Run Start Date/Time Date/Time
- 1 - If
- 2 - Update Record
 - Incident WorkFlow Record Record
 - Incident WorkFlow Table Table
 - Action Status Object
- 3 - Update Record
 - Incident WorkFlow Record Record

Branch 3 Action — Updating Incident WorkFlow Record to Category = Access, Subcategory = Forgot Password

Auto Classify School IT Tickets Inactive

6 Else If SD is Forgot password

7 * then Update Incident WorkFlow Record

8 Else If

Condition Label: SD is Performance

* Condition 1: Trigger... Short Descri... X

containing Slow Computer

or and

Add another condition set(OR)

Delete Cancel Done

Status: Modified Application: Global

Data Collapse All

- Incident WorkFlow Record Record
- Incident WorkFlow Table Table
- Run Start Time UTC Date/Time
- Run Start Date/Time Date/Time
- 1 - If
- 2 - Update Record
 - Incident WorkFlow Record Record
 - Incident WorkFlow Table Table
 - Action Status Object
- 3 - Update Record
 - Incident WorkFlow Record Record

Branch 4 (Else If) — Evaluating Short Description Contains 'Slow Computer'

Auto Classify School IT Tickets Inactive

9 then Update Incident WorkFlow Record

Action Properties

Action: Update Record

Action Inputs

* Record: Trigger... Incident WorkFlow...

* Table: Incident WorkFlow [u_incident_w...]

* Fields: Category: Performance, Subcategory: Slow Computer

+ Add field value

Data Collapse All

- Incident WorkFlow Record: Record
- Incident WorkFlow Table: Table
- Run Start Time UTC: Date/Time
- Run Start Date/Time: Date/Time

1 - If

2 - Update Record

- Incident WorkFlow Record: Record
- Incident WorkFlow Table: Table
- Action Status: Object

3 - Update Record

- Incident WorkFlow Record: Record

Branch 4 Action — Updating Incident WorkFlow Record to Category = Performance, Subcategory = Slow Computer

Auto Classify School IT Tickets Inactive

Target Record: Trigger... Incident WorkFlow...

Table: Incident WorkFlow [u_incident_w...]

Include Watermark: ☒

* To: Trigger - Record C... Em...

CC:

BCC:

* Subject: Your Request for the issue has been submitted

Body: Ticket confirmation message

Status: Modified Application: Global

Data Collapse All

- Action Status: Object

5 - Else If

No data available

6 - Update Record

- Incident WorkFlow Record: Record
- Incident WorkFlow Table: Table
- Action Status: Object

7 - Else If

No data available

8 - Update Record

- Incident WorkFlow Record: Record

Step 9 Action — Configuring 'Send Email' to Caller Email with Subject and Confirmation Body

